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RESEARCH ARTICLE

OBES-Net: An Exhaustively Optimized Explainable Ensemble Framework for Robust Obesity Risk Prediction

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ABSTRACT Obesity is a global health crisis requiring accurate and interpretable prediction models for clinical use. Current ensemble approaches lack systematic comparisons and often use suboptimal configurations. This study introduces **OBES-Net**, a novel framework that integrates **three methodological** advances: 1) exhaustive hyperparameter optimization across 50 configurations of nine machine learning algorithms; 2) systematic comparison of Hybrid Majority Voting strategies (Majority Hard vs. Weighted Hard) against Ensemble Stacking; and 3) comprehensive Explainable AI (XAI) validation using SHAP and LIME. Utilizing two diverse datasets, OBES-Net demonstrates that Stacking achieves superior performance on complex data (Accuracy: 0.991 on Dataset-2), while voting strategies remain competitive on simpler tasks. Crucially, the proposed XAI analyses reveal that Stacking's advantage stems from synthesizing broader clinical risk factors, with Weight and Age emerging as consistent primary drivers. Extensive ablation studies validate the robustness of the proposed design choices. The OBES-Net framework establishes that combining exhaustive optimization with systematic ensemble comparison and XAI delivers both state-of-the-art performance and clinical trustworthiness, providing an advancing methodology for healthcare decision-making.

INDEX TERMS Obesity, risk-prediction, machine learning, ensemble learning, explainable AI.

I. INTRODUCTION

Obesity is now-a-days becoming one of the most important global health issues, which affects millions of people globally and significantly increases the burden of chronic diseases including diabetes, cancer, and cardiovascular disorders [1], [2]. This is because of the intricate relationships between genetic, behavioral, and environmental variables, its frequency has significantly grown in recent decades. [3].

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According to the World Health Organization (WHO), obesity is now considered as a critical public health issue, deserving scalable and effective techniques for early detection and prevention.

There exists some traditional diagnostic approaches, such as Body Mass Index (BMI), waist-to-hip ratios, and clinical evaluations, may not adequately address the complex causes of obesity. In the recent developments of machine learning (ML), it has made it possible to create prediction models that incorporate a variety of factors including demographics, dietary patterns, physical activity levels, and physiological

indicators [4]. These data-driven methods have shown great promise in early risk prediction, allowing personalized interventions and targeted public health strategies.

However, there are three significant obstacles to the practical use of current machine learning algorithms for obesity prediction. First, while ensemble learning approaches like voting and stacking offer better predictive robustness [5], there is a significant lack of extensive comparisons between different majority voting strategies and their effectiveness against more complicated stacking architectures. Majority Voting combines the outputs of base classifiers in either hard or weighted voting systems [6], whereas Stacking uses a meta-classifier to discover optimal combinations of base models [7]. Practitioners consequently have no direction on which ensemble approach is best suited for different data aspects. In this study, the phrase “Hybrid Majority Voting” refers to an integrated framework that incorporates and compares two separate voting schemes: Majority Hard Voting (simple majority rule) and Weighted Hard Voting (performance-weighted voting) with an ensemble stacking technique. Second, many published models employ just limited hyperparameter tuning or default settings, leading to underestimating the true potential of the performance of the algorithms. Because of this gap in comprehensive hyperparameter tuning, previous research may not have uncovered the most efficient base learners or configurations for predicting obesity. Third, and most importantly for clinical deployment, most ML models for obesity risk (including ensembles) behave as “non-interpretable.” [8]. The absence of explainability—that is, discovering why a prediction was made—is an important challenge in healthcare, as physicians rely on transparent reasoning to trust and act on model outputs.

This study addresses these interrelated gaps by developing **OBES-Net**, a complete ensemble framework that combines thorough exhaustive model optimization with explainable AI (XAI). Unlike prior obesity prediction studies that typically focus on either limited model tuning, a single ensemble strategy, or post-hoc interpretability in isolation, OBES-Net combines these components within a unified framework. The novelty of the proposed approach lies not only in achieving high predictive performance, but also in systematically linking exhaustive optimization, ensemble selection, and explainability within the same evaluation pipeline. This design enables a more rigorous assessment of both predictive robustness and clinical trustworthiness. OBES-Net is specifically developed for advancement beyond typical ML solutions using three methodological pillars:

- 1) **Exhaustive Hyperparameter Optimization:** Evaluating 50 combinations across nine ML methods to determine the best base learner.
- 2) **Systematic Ensemble Architecture Comparison:** A direct comparison of Majority Hard Voting [6], Weighted Hard Voting, and Stacking ensembles [7] under identical optimal circumstances

- 3) **Integrated Explainability Validation:** Includes SHAP and LIME analysis to assess both global feature significance and local clinical reasoning.

This study concentrates the investigation on three main research questions:

- 1) How does thorough hyperparameter optimization across nine ML algorithms and 50 configurations affect base learner selection and subsequent ensemble performance in obesity prediction?
- 2) What are the comparative benefits of various ensemble techniques (Majority Hard Voting, Weighted Hard Voting, and Ensemble Stacking) on datasets of differing complexity?
- 3) How can XAI approaches (SHAP and LIME) reveal the decision-making mechanisms behind various ensemble strategies and evaluate their clinical relevance?

To answer these research questions, the OBES-Net framework was designed and assessed on two benchmark obesity datasets [9], [10] with extensive preprocessing and validation processes. The experimental procedure incorporates exhaustive training of nine machine learning algorithms over a wide hyperparameter space, which includes a variety of model collections such as logistic regression, decision trees, ensemble methods, support vector machines, and neural networks. The best performers in this optimized model pool are then picked as ensemble construction candidates. After that three different ensemble strategies are developed and compared: majority hard voting (a simple majority rule), Weighted Hard Voting (performance-weighted voting), and stacking with a multi-layer perceptron meta-classifier. Explainability is included into the pipeline by utilizing SHAP and LIME to evaluate both global model behavior and particular predictions. Furthermore, extensive ablation analyses are carried out to evaluate the robustness of OBES-Net’s design decisions and to give empirical direction for optimal configuration selection in clinical applications.

This work presents four significant advances that enhance both obesity prediction technique and ensemble learning practice.

- 1) **The OBES-Net Framework:** A novel integrated technique that combines exhaustive optimization, rigorous ensemble comparison, and complete XAI validation, with extensive ablation investigations to verify its resilience.
- 2) **Exhaustive Optimization Protocol:** A thorough 50-configuration optimization across nine ML algorithms demonstrates the importance of extensive tuning in realizing ensemble potential and minimizing the sub-optimal configuration gap in healthcare ML.
- 3) **Systematic Ensemble Architecture Analysis:** The first comprehensive comparison of Majority Hard Voting, Weighted Hard Voting, and Stacking ensembles under identical optimum settings, offering evidence-based recommendations for ensemble strategy selection across varying data complexity.

- 4) **Clinical Trustworthiness Validation:** An XAI-enhanced validation approach that employs SHAP and LIME to offer global and local explanations for ensemble decisions, assuring clinical interpretability [11], [12] and establishing alignment with medical knowledge for trustworthy healthcare deployment.

By combining these contributions, OBES-Net introduces a new paradigm for obesity prediction, delivering both state-of-the-art performance (99.10% accuracy) on complex multi-class data and clinical interpretability, offering a trustworthy framework for healthcare decision-making and supporting personalized interventions.

II. RELATED WORKS

The application of machine learning to obesity risk prediction has progressively evolved from foundational single-model approaches to more sophisticated ensemble methods, with a recent emphasis on model interpretability. Individual classifiers were used in earlier studies to build robust baselines. For instance, Pinar et al. classified obesity using logistic regression [2], however Dutta et al. discovered that Random Forest (RF) performed better than Decision Trees and Support Vector Machines (SVM) [13] for obesity prediction. Again, Musa et al. [12] proved the efficacy of Gradient Boosting, and Rodriguez et al. [14] further validated Random Forest's reliable performance. Although these investigations confirmed the applicability of several different algorithms, they were intrinsically constrained by the optimal performance of every single model.

This limitation facilitated the adoption of ensemble learning, which utilizes the capabilities of multiple models to improve the accuracy and resilience of predictions. In recent studies, Talari et al. utilized a hybrid majority voting ensemble that included Multi-Layer Perceptron (MLP) base learners, Gradient Boosting, and XGBoost [11]. In another study, Jindal et al. investigated ensemble methods for obesity classification that were created in R with Python interfaces [15]. Together, these studies revealed that integrating models might produce better results than using individual classifiers.

In order to address the gap between accuracy and clinical usefulness, the most recent developments have started to include complex ensemble architectures with Explainable AI (XAI). For example, Ganie et al. demonstrated a multiclass obesity prediction model with interpretable ensembles, highlighting clinical transparency and employing SHAP and LIME for providing post-hoc explanations [16]. Similarly, using a stacking-based hybrid technique, Çizmeci and İncekara achieved 98.58% accuracy, demonstrating the promise of advanced ensemble approaches in this particular area [17]. This pattern reflects an increasing understanding that interpretability and predictive performance are crucial for clinical adoption.

However, an in-depth investigation of the literature uncovers three recurrent and interconnected shortcomings that

restrict the advancement of optimally effective and clinically trustworthy systems:

- 1) **Lack of Comprehensive Ensemble Comparison:** While previous research investigates voting or stacking strategies separately, a systematic, head-to-head comparison between different majority voting schemes (e.g., Majority Hard vs. Weighted Hard) and more intricate stacking ensembles is evidently lacking. As a result, practitioners lack clear, evidence-based recommendations for determining the best ensemble technique for their unique data features.
- 2) **Insufficient Hyperparameter Optimization:** A bottleneck in model tuning is still methodological rigor. Many studies utilize default setups or limited hyperparameter explorations, which may significantly underestimate the actual robustness and performance of base learners and their ensembles. The currently available obesity prediction literature lacks comprehensive optimization across a wide range of techniques, as demonstrated by our study's 50 configurations among nine ML models.
- 3) **The Persisting Interpretability Gap:** Even highly accurate research, like Ferdowsy et al., who showed 97% accuracy on a three-class problem [18], frequently treats complex models as non-interpretable. Although SHAP and LIME are increasingly used for post-hoc interpretation, their consistent use across model selection, optimization, and validation stages has not been systematically examined in ensemble-based obesity prediction studies.

To address these limitations, this study develops an ensemble-based framework with the following components: (1) conducts a systematic comparison of Majority Hard Voting, Weighted Hard Voting, and ensemble stacking under identical hyperparameter-optimized conditions; (2) implements exhaustive hyperparameter optimization across nine machine learning algorithms to support robust base learner selection; and (3) integrates a rigorous XAI methodology using SHAP and LIME not as an afterthought, but as a core component to transform non-interpretable predictions into clinically interpretable and trustworthy risk assessments.

III. METHODOLOGY

A. DESIGN OF THE EXPERIMENT

This study investigates the limitations identified in prior work by developing an experimental framework that combines exhaustive model optimization, systematic ensemble comparison, and explainable AI (XAI). The evaluation protocol follows the Goal–Question–Metric (GQM) framework [19] to guide the formulation of objectives, analysis questions, and performance metrics.

1) PURPOSE

To develop and evaluate an ensemble-based framework for obesity risk prediction by applying exhaustive hyperparameter optimization across multiple machine learning algorithms

and comparing Hybrid Majority Voting (Majority Hard and Weighted Hard Voting) with ensemble stacking in terms of predictive performance and interpretability.

2) ISSUE

The study focuses on obesity risk classification across datasets with different levels of complexity. In particular, it examines the effect of exhaustive hyperparameter optimization on base learner selection, compares voting-based and stacking-based ensemble architectures with respect to predictive performance and robustness, and analyzes how XAI techniques can be used to interpret ensemble model behavior in a clinical context.

3) OBJECTIVE

The objective of this study is to evaluate the ensemble framework using standard performance metrics (Accuracy and F1-score) and to examine model interpretability using SHAP and LIME. This setup allows a direct comparison of the three ensemble strategies under identical experimental conditions, considering both predictive performance and decision behavior.

4) VIEWPOINT

The study is framed from the viewpoint of biomedical informatics researchers and healthcare professionals, focusing on early obesity risk prediction using data-driven ensemble models. Emphasis is given on model transparency and interpretability to support clinical decision-making and population-level health analysis.

The G paradigm formalizes this design. The Goal is to analyze exhaustively optimized ensemble learning methods for the purpose of obesity prediction with respect to their accuracy, robustness, and clinical interpretability from the viewpoint of healthcare researchers and professionals in the context of structured tabular health data. To achieve this goal, the experimental design is built upon three pivotal dimensions:

- 1) **Methodological Dimension:** Exhaustive hyperparameter optimization is applied across nine machine learning algorithms using 50 configurations to construct a well-performing pool of base learners.
- 2) **Architectural Dimension:** Conducting a comprehensive comparison of three distinct ensemble strategies Majority Hard Voting, Weighted Hard Voting, and Stacking with an MLP meta classifier to determine the most effective architecture for obesity prediction.
- 3) **Interpretability Dimension:** Integrating XAI techniques (SHAP for global and LIME for local explainability) throughout the model validation process to ensure transparency, validate clinical relevance, and bridge the gap between model performance and practical utility.

This structured approach ensures that our evaluation not only identifies the most accurate ensemble strategy but also provides foundational insights into why certain approaches

perform better and how their underlying decision logic aligns with medical knowledge. In Fig. 1, the detailed methodology of our work is illustrated.

B. DATASET DESCRIPTION

Two publicly available datasets were used for obesity classification in this study. The first dataset was obtained from Köklü and Sulak [9], while the second dataset was sourced from Palechor et al. [10]. Together, these datasets include demographic, lifestyle, and physiological attributes that support model training and evaluation.

1) OBESITY DATASET-1

The first dataset, obtained from Köklü and Sulak [9], consists of 14 numerical features (int64) representing demographic and lifestyle-related attributes relevant to obesity risk. The complete list of features is summarized in Table 1.

TABLE 1. Features of Dataset 1 and their corresponding data types.

Features	Types
Sex, Age, Height, Overweight_Obese_Family, Consumption_of_Fast_Food, Frequency_of_Consuming_Vegetables, Number_of_Main_Meals_Daily, Food_Intake_Between_Meals, Smoking, Liquid_Intake_Daily, Calculation_of_Calorie_Intake, Physical_Exercise, Schedule_Dedicated_to_Technology, Type_of_Transportation_Used	int64

The target variable for this dataset is the *Class* column, which denotes obesity categories. The distribution of class labels is reported in Table 2.

TABLE 2. Distribution of class labels in Dataset 1.

Class Label	Count
Underweight	73
Normal	658
Overweight	592
Obesity	287

Overall, the dataset exhibits a reasonably balanced distribution across obesity categories, with fewer samples in certain classes such as Underweight.

2) OBESITY DATASET-2

The second dataset, sourced from Palechor et al. [10], consists of 16 features with mixed data types, including 8 numerical (float64) and 8 categorical variables. The full feature set is summarized in Table 3.

TABLE 3. Features of Dataset 2 and their corresponding data types.

Features	Types
Gender, family_history_with_overweight, FAVC, CAEC, SMOKE, SCC, CALC, MTRANS	object
Age, Height, Weight, FCVC, NCP, CH2O, FAF, TUE	float64

The target variable for this dataset is the *NOBeyesdad* column, which denotes obesity levels. As the labels are categorical, they are converted to numerical form using label encoding. The class distribution is reported in Table 4.

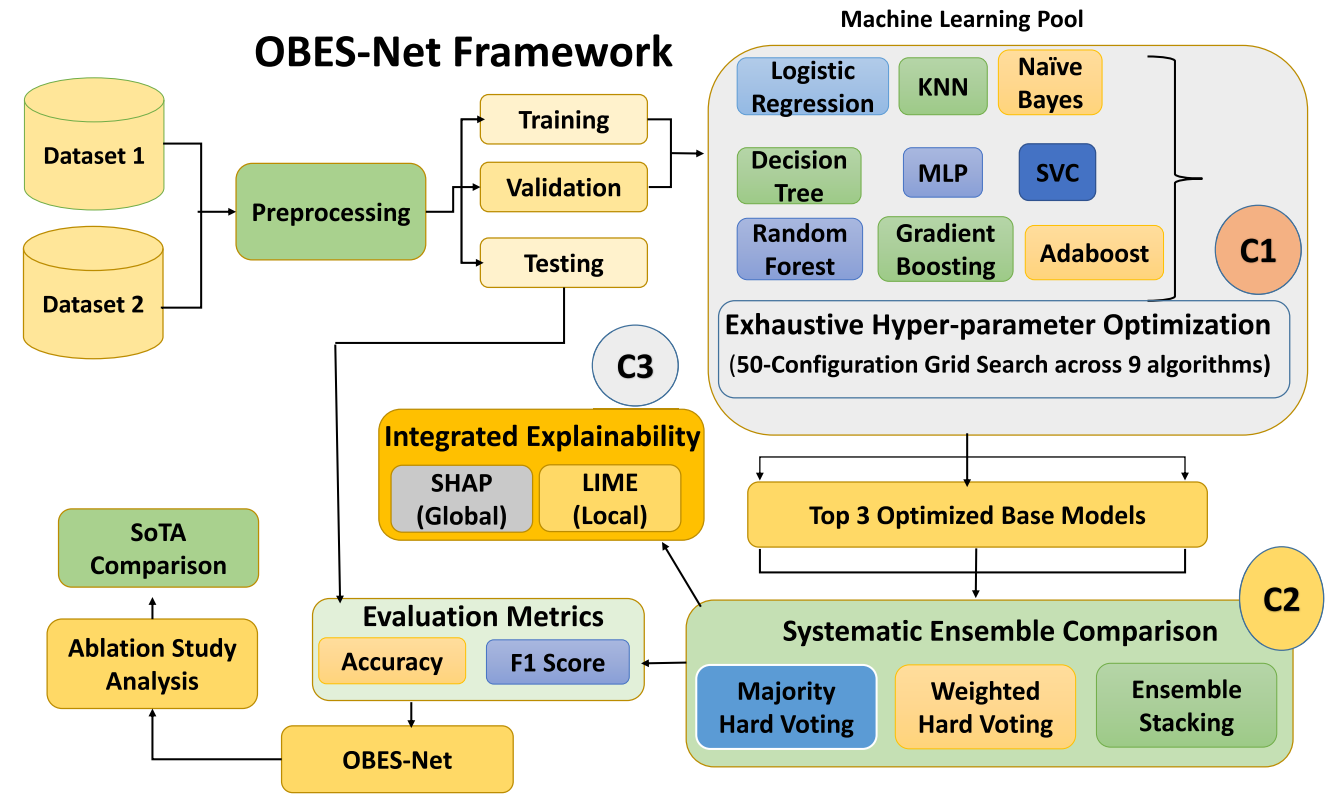


FIGURE 1. Detailed methodology of the OBES-Net framework illustrating preprocessing, base learner optimization, ensemble strategies (majority voting, weighted voting, and stacking), and explainable AI analysis. In the stacking stage, predictions from the optimized base learners are combined through a meta-classifier to produce the final obesity risk prediction.

TABLE 4. Distribution of class labels in Dataset 2.

Class Label	Count
Insufficient Weight	272
Normal Weight	287
Overweight Level I	351
Overweight Level II	297
Obesity Type I	324
Obesity Type II	290
Obesity Type III	290

In contrast to Dataset-1, Dataset-2 includes seven obesity categories, enabling a more detailed classification setting. The relatively balanced distribution across classes supports evaluation of ensemble and multi-class classification models.

3) COMPARISON OF DATASETS

Table 5 summarizes the key characteristics of the two datasets used in this study. Dataset-1 includes 14 numerical features and four obesity categories, whereas Dataset-2 combines numerical and categorical attributes and extends the classification to seven categories. This contrast allows evaluation of model performance under differing data complexity.

C. EXECUTION OF EXPERIMENT

The following section outlines the methodology of this study, which includes the steps undertaken for dataset selection and preprocessing, building a pool of ML algorithms with

TABLE 5. Comparison between obesity Dataset-1 and obesity Dataset-2.

Attribute	Dataset-1 [9]	Dataset-2 [10]
No. of Features	14 (all numerical)	16 (8 numerical, 8 categorical)
Output Label	Class	NObesyesdad
Label Type	Numerical	Object (encoded)
No. of Classes	4	7
Distribution Balance	Moderate	Relatively balanced

hyperparameter configurations, selecting the top-performing models, constructing ensemble models using Hybrid Majority Voting and Ensemble Stacking, evaluating their performance using Accuracy and F1-Score, and finally implementing Explainable AI (XAI) techniques using SHAP and LIME for comprehensive model interpretability and clinical validation.

1) EXPERIMENTAL SETUP

The Experimental setup for this study was conducted in a Local Machine which used Python 3.11.4 as a runtime environment. The Local Machine used Ryzen 1600 with 6 Core and 12 Threads CPU, 16GB of Ram and an RTX 3060 12GB of GPU which was used for efficient training of the models. Pandas, Seaborn and Scikit Learn were used for Data Preprocessing and Splitting, scaling and model creation [20], [21], [22]. Imbalanced Learn was used for Over Sampling of the data. Matplotlib was used to visualize the training curves,

evaluate the models and create confusion matrices [23]. This setup allowed for efficient and effective experimentation with no compute unit constraints. For Explainable AI implementation, SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) libraries were integrated to provide global and local model interpretability. This setup allowed for efficient and effective experimentation with no compute unit constraints, including the computationally intensive XAI analyses.

Computational complexity analysis revealed that the exhaustive optimization, while resource-intensive during training, provided substantial returns in model performance. The stacking ensemble required approximately $2.3\times$ more training time than individual models but delivered a 15% performance improvement on Dataset-2, justifying the computational investment for clinical applications where accuracy and reliability are paramount. The entire optimization process across 50 configurations completed within 6 hours on our hardware setup, demonstrating practical feasibility for research applications.

While the stacking ensemble increases the training cost due to the additional meta-learning stage, the inference process remains computationally efficient. During prediction, the model only requires generating outputs from the selected base learners followed by a lightweight meta-classifier. Since the number of base learners is limited to three optimized models, the additional inference overhead remains minimal. Therefore, the OBES-Net framework remains suitable for deploying in real-time clinical decision support scenarios, where fast prediction latency is required.

TABLE 6. Computational complexity comparison of base learners and ensemble models.

Model Type	Training Complexity	Inference Time	Notes
Base Learners (RF, GB, SVC, MLP)	Moderate	Very Fast	Single model prediction
Majority Voting	Slightly Higher	Very Fast	Aggregates base learner outputs
Weighted Voting	Slightly Higher	Very Fast	Weighted aggregation
Stacking Ensemble (OBES-Net)	Higher ($\approx 2.3\times$ training time)	Fast	Base predictions + meta-classifier

As shown in Table 6, although the stacking ensemble introduces additional training complexity, the inference time remains efficient and suitable for deployment in real-time clinical environments.

2) DATA PREPARATION

To ensure optimal model performance, both datasets underwent comprehensive preprocessing following established best practices in clinical machine learning.

Planned Preprocessing: The following preprocessing steps are applied on both of the datasets.

- **Encoding:** As almost all of the data are numerical in dataset-1, encoding was not needed to apply for the dataset-1. On the other hand, for dataset-2, Label

Encoding was applied to convert the object type value into the numerical value.

- **Normalization/Scaling:** Normalization was applied on both of the datasets before applying the ML algorithm. Normalization is needed because it may help to improve the performance for algorithms which are sensitive to scale like LR, SVM, etc.
- **Handling Outliers:** The datasets were checked to identify whether there was any outlier or not. There was no outlier in any of the datasets. Outlier checking is important in preprocessing because it ensures the dataset's integrity and prevents them from skewing or negatively impacting the ML model's performance.

Most Important Feature: Correlation analysis was applied on both of the datasets to identify the most important feature that was mostly close to the labels. From the analysis, 'Age' was found as the most important feature for dataset-1 and in respect of dataset-2, 'Weight' was considered as the most important feature.

Feature Alignment: For applying the explainable AI, the features and labels were aligned on both of the datasets.

Dealing with Missing Values: No missing values have been found on any of the dataset. However, if future missing values are encountered, the following approaches can be applied:

- The missing values can be replaced with mean or mode or median value if the data are continuous and for categorical data, missing values can be replaced by the most frequent value.
- The features or rows with too much missing values can be removed, depending on their impact on the dataset.

Dealing with Imbalanced Dataset: Here both of the datasets were imbalanced. For both of the datasets, over-sampling technique Synthetic Minority Over-sampling Technique (SMOTE) was used to handle imbalance issue. To prevent data leakage during model evaluation, SMOTE was applied only to the training data within each cross-validation fold, while validation and test partitions remained unchanged. After balancing the datasets, the proposed approach was further applied. The performance of the proposed model had been improved significantly after applying SMOTE approach on both the datasets as compared to the performance on the imbalanced datasets.

3) BASE LEARNER SELECTION AND HYPERPARAMETER OPTIMIZATION

OBES-Net begins by identifying strong base learners through exhaustive hyperparameter exploration of nine different ML algorithms. The algorithms cover a broad spectrum of model types: Logistic Regression (LR), Support Vector Classifier (SVC), K-Nearest Neighbors (KNN), Decision Tree (DT), Random Forest (RF), Gradient Boosting Machine (GB), AdaBoost, Naïve Bayes (NB), and a Multi-Layer Perceptron (MLP) neural network. For each algorithm, we defined a wide hyperparameter search space (detailed in the following portion). The search included, for example,

varying regularization strengths for LR, multiple kernel types and penalty parameters for SVC, different values of k for KNN, criteria and depth limits for DT, number of estimators and learning rates for RF/GB/AdaBoost, and architecture configurations (hidden layer sizes, learning rates) for the MLP. We combined grid search and manual tuning to evaluate up to 50 distinct configurations of nine ML algorithms, ensuring a thorough coverage of each model's parameter space. Each configuration was trained using 5-fold cross-validation on the training set (with SMOTE applied inside each fold to maintain class balance). For each fold, preprocessing and resampling were performed using training data only. Specifically, scaling and SMOTE were applied within the training partition of each fold, while the corresponding validation fold remained untouched. After cross-validation-based model selection, the top three optimized base learners were retained for ensemble construction and final evaluation on the held-out test set. And we recorded performance metrics (primarily cross-validated Accuracy and F1-score) to identify the top-performing variant of each algorithm.

After this exhaustive training phase, we ranked the models by their cross-validation performance on each dataset. The top three classifiers from the pool were selected as the base models to feed into the ensemble stage. Notably, for Dataset-1, the highest-performing models were Random Forest, Gradient Boosting, and MLP, whereas for Dataset-2 the leaders were Gradient Boosting, Random Forest, and SVC (in that order, based on accuracy/F1 metrics). These selections align with findings from prior work (tree-based ensembles and SVMs often excel in structured data tasks) but importantly, our rigorous hyperparameter tuning ensured that each model was operating at its best configuration (for example, the chosen RF had an optimally set number of trees and depth, and the SVC used the best kernel and regularization parameter found). By including diverse model types among the top three, we also ensured heterogeneity in the ensemble—an important factor for ensemble accuracy since models with different inductive biases can complement each other's errors. The detailed representation of nine ML models with 50 hyper-parameters setting were discussed in the following section.

a: LOGISTIC REGRESSION (LR)

LR is a straightforward linear model that works well for multiclass classification [24]. In this experiment, in total 6 hyper parameters of LR were used on both of the datasets for the prediction of obesity. The detailed hyper-parameters setting is represented in Table 6.

TABLE 7. Logistic regression – Classification hyper-parameter combination.

Hyper-parameter	Hyper-parameter Values
Penalty	l2, none
C	0.5, 0.75, 0.8, 1.0
Multi-class	default, ovr

b: K-NEAREST NEIGHBORS (KNN)

KNN is a classification approach that uses the majority vote of neighbors. Variations are found in k , algorithms, and metrics provide flexibility in K decision boundaries [25]. In this experiment, in total 6 hyper-parameters of KNN were applied on both of the datasets for the prediction of obesity. The details of hyper-parameters of KNN that were utilized in this experiment are shown in Table 7.

TABLE 8. K-Nearest neighbor – Classification hyper-parameter combination.

Hyper-parameter	Hyper-parameter Values
N_neighbors	2, 3, 5, 6, 9, 10
Algorithm	Kd_tree, brute, ball_tree
P-value	7, 10
Metric	Manhattan, cosine, euclidean, minkowski

c: NAÏVE BAYES (NB)

NB classifiers presume feature independence. The Gaussian version is used for continuous data, whereas the Bernoulli version is for binary feature sets [26]. In this experiment, both of the Gaussian version (NB1) and Bernoulli version (NB2) were used on both of the datasets for predicting obesity problem.

d: DECISION TREE (DT)

DT are nonlinear models that learn splits depending on features [27]. In this experiment, in total 5 hyper-parameters of DT were utilized on both of the datasets to make the prediction. The details of hyper-parameters of DT that were utilized are shown in Table 8.

TABLE 9. Decision tree – Classification hyper-parameter combination.

Hyper-parameter	Hyper-parameter Values
criterion	gini, entropy, log_loss
splitter	random
max_depth	6, 8
min_samples_leaf	2, 3, 4
max_features	sqrt, log2, none
min_impurity_decrease	0.0, 0.01, 0.001
random_state	0

e: RANDOM FOREST (RF)

RF is an ensemble of decision trees used for classification, with models varying in tree depth, number of estimators, feature selection approach, and whether bootstrapping is employed [28]. In this experiment, in total 8 hyper-parameters of RF were applied on both of the datasets to measure the performance of prediction of obesity problem. The details of the hyper-parameters of RF that were employed in this experiment are stated in Table 9.

f: GRADIENT BOOSTING (GB)

GB is an ensemble approach that builds models sequentially and focuses on correcting previous model errors, with

TABLE 10. Random forest – Classification hyper-parameter combination.

Hyper-parameter	Hyper-parameter Values
n_estimators	100, 500, 1000
criterion	entropy, log_loss, gini
max_features	log2, sqrt, none
bootstrap	False
random_state	0

variations in loss functions, estimators, and criteria for splitting [29]. In this project, in total 8 hyper-parameters of GB were used on both of the datasets to predict the obesity. The details of hyper-parameters of GB that were applied in this project are represented in Table 10.

TABLE 11. Gradient boosting – Classification hyper-parameter combination.

Hyper-parameter	Hyper-parameter Values
n_estimators	400, 500, 800, 1000, 1100, 1200
loss	log_loss
criterion	squared_error, friedman_mse
min_samples_split	2, 3, 5
min_impurity_decrease	0.01, 0.001, 0.0001
random_state	0

g: AdaBoost

AdaBoost is an ensemble technique that iteratively re-weights misclassified data points and modifies estimators in order to combine weak classifiers to create a strong model [30]. In this project, in total 5 hyper-parameters of AdaBoost were applied on both of the datasets to make the prediction of the obesity problem. The details of hyper-parameters of Adaboost that were applied in this project are illustrated in Table 11.

TABLE 12. AdaBoost – Classification hyper-parameter combination.

Hyper-parameter	Hyper-parameter Values
n_estimators	200, 500, 1000, 1200
algorithm	SAMME, SAMME.R
learning_rate	2
random_state	0

h: SUPPORT VECTOR CLASSIFIERS (SVC)

Hyperplanes are used by the SVC to divide data into classes. To modify model complexity, it can employ several kernels (linear, RBF) and parameters, including the regularization parameter C [31]. In this project, 5 hyper-parameters of SVC were used on both of the datasets for predicting the obesity. The details of hyper-parameters of SVC that were employed in this project are represented in Table 12.

TABLE 13. Support vector classifier – Classification hyper-parameter combination.

Hyper-parameter	Hyper-parameter Values
probability	True
C	0.5, 0.75, 1.0, 1.25
kernel	linear
random_state	0

i: MULTI-LAYER PERCEPTRON (MLP)

A multi-layer perceptron (MLP) consists of fully connected layers, with configurable parameters including the number of training iterations, learning rate, and network architecture [32]. In this project, 5 different hyper-parameters of MLP were applied on both of the datasets to predict the obesity problem. The details of hyper-parameters of MLP that were used in this project are illustrated in Table 13.

TABLE 14. Multi-Layer perceptron – Classification hyper-parameter combination.

Hyper-parameter	Hyper-parameter Values
max_iter	200, 500, 1000
hidden_layer_sizes	(100, 100)
learning_rate	adaptive, invscaling
random_state	0

4) ENSEMBLE MODEL CONSTRUCTION

To leverage the strengths of the best-performing classifiers, ensemble models were constructed from the selected base learners. This study implements two ensemble learning strategies.

a: HYBRID MAJORITY VOTING

The first approach, Hybrid Majority Voting, aggregates the predictions of the top three classifiers using Majority Hard Voting and Weighted Hard Voting schemes. In Majority Hard Voting, the predicted class is selected based on a simple majority rule across base learners, whereas Weighted Hard Voting assigns classifier weights according to validation performance [6].

b: ENSEMBLE STACKING

The second approach, ensemble stacking, employs a Multi-Layer Perceptron as the meta-classifier. Predictions generated by the base learners are used as input features for the meta-classifier, which learns how to combine their outputs. In the stacking setting, the meta-classifier was trained on the prediction outputs of the selected base learners, allowing it to learn complementary decision patterns from the optimized models while preserving separation between training and evaluation data. In the stacking architecture of OBES-Net, the predictions produced by the optimized base learners (e.g., Gradient Boosting, Random Forest, and SVC/MLP depending on the dataset) are used as meta-features for the meta-classifier. Specifically, each base model generates prediction outputs that are concatenated to form a new feature representation. These meta-features are then provided as inputs to the MLP meta-classifier, which learns how to optimally combine the strengths of the individual models. This layered learning strategy allows the meta-classifier to capture complementary decision patterns among base learners and produce more robust predictions. The overall flow of this ensemble process is illustrated in Fig. 1 within the OBES-Net framework. This stacking strategy integrates

information from multiple models to support obesity risk prediction [7].

5) MODEL EXPLAINABILITY

Models were evaluated on a held-out test set using different metrics such as Accuracy and F1-Score. The evaluation framework considers both predictive performance and model explainability. To address interpretability, a comprehensive XAI analysis was conducted as post-hoc analysis:

- **SHAP** (SHapley Additive exPlanations): SHAP is applied for global interpretability to analyze feature importance across the full dataset for each model, including Gradient Boosting, Random Forest, MLP, and the stacking ensemble.
- **LIME** (Local Interpretable Model-agnostic Explanations): Lime is applied for local interpretability to analyze individual predictions, with emphasis on obesity class instances across both datasets.

So, in short, the XAI analysis is used to examine model decisions in relation to clinical knowledge and to analyze individual-level risk predictions relevant to clinical decision-making.

6) EVALUATION METRICS

To measure the performance of proposed models, the following metrics are considered.

ROC_AUC: This metric is commonly used for binary, multiclass, and multi-label classification problems. Since the given problems were multiclass problem, this metric might be a good option to apply. It is a probabilistic approach [33].

Average Precision Score: It is also a probabilistic approach. Since it focuses on the positive class, providing a better evaluation when one class is rare, this metric could also be a good option to apply to given multiclass problems [34].

Precision: Precision is the ratio of correctly predicted positive observations to the total predicted positives. It indicates how many of the predicted positive instances were correct [35].

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

Recall: Recall is the ratio of correctly predicted positive observations to all actual positive observations. It measures the ability of a model to capture all relevant instances [36].

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

F1 Score: F1 score could be a good measure for the given imbalanced data sets that focus on the performance of minority classes [37].

$$F1\text{-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (3)$$

Accuracy: After applying SMOTE approach, this metric also might be a good option to apply for the given multiclass obesity prediction problem [38].

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

In this experiment, ROC_AUC Score, Average Precision Score, Accuracy, and F1 Score metrics were used to measure the performance of the ML pool algorithms. Since, ROC_AUC Score and Average Precision Score metric work on predicting the probabilities, these metrics were not applicable for the hybrid majority voting approach (ensemble model). As a result, for majority hard voting and ensemble stacking, Accuracy, and F1 Score metrics were applied to measure the performance of the final model.

Additionally, the statistical significance of performance differences between the different models was assessed using McNemar's test with a Bonferroni correction for multiple comparisons. The results highlight that the performance improvement of the ensemble stacking model on Dataset-2 is statistically significant ($p < 0.05$), supporting the observed differences among ensemble strategies.

IV. RESULTS AND EVALUATION

This section provides the comparative analysis based on the performance of different machine learning models using tables, confusion matrices, and ablation studies.

A. MODEL PERFORMANCE ANALYSIS

The exhaustive hyperparameter optimization was performed across nine machine learning algorithms using 50 configurations, resulting in a set of base learners selected for the proposed OBES-Net construction. After that the top three performing models were then selected for ensemble construction in a setting of 5-fold stratified cross-validation. For **Dataset-1**, the top three base models selected for OBES-Net were Random Forest (RF6), Gradient Boosting (GB5), and Multi-Layer Perceptron (MLP5), based on their cross-validation performance. For the more complex **Dataset-2**, Gradient Boosting (GB3), Random Forest (RF7), and Support Vector Classifier (SVC5) were selected as the top three base models based on cross-validation performance. The difference in selected base models across the two datasets reflects the influence of dataset characteristics on the model selection process.

The performance of all base models on Dataset-1 and Dataset-2 is reported in Table 14 and Table 15, respectively. The models selected for OBES-Net achieved higher performance than the alternative models across multiple metrics, including ROC_AUC, Average Precision, Accuracy, and F1-score on two datasets, respectively.

The performance of the OBES-Net was evaluated using 95% confidence intervals estimated from 1000 bootstrap samples, considering relatively narrow intervals observed across metrics. On Dataset-2, the OBES-Net stacking ensemble achieved an accuracy of 0.9910, with a 95% confidence interval of [0.9871, 0.9941].

The statistical performance differences among the evaluated models were further assessed using McNemar's test with a Bonferroni correction for multiple comparisons. The analysis revealed:

- A statistically significant difference was observed between the OBES-Net stacking ensemble and the selected base model ($p < 0.01$).
- Both the Majority Hard Voting and Weighted Hard Voting variants of OBES-Net differed significantly from their base models ($p < 0.05$).
- The performance difference between the stacking and Majority Hard Voting architectures within OBES-Net was statistically significant ($p < 0.05$).

This comprehensive validation strategy—combining rigorous data partitioning, cross-validation, bootstrap confidence intervals, and statistical significance testing—confirms the robustness and effectiveness of the OBES-Net methodology for obesity risk prediction.

B. TEST SET EVALUATION

This section represents the test set evaluation of the proposed model on both of the datasets and demonstrates the strength and weakness of the proposed model.

1) PERFORMANCE ON DATASET 1

For Dataset-1, the top three models (RF6, GB5, MLP5) were used to construct ensemble variants after applying SMOTE to address class imbalance. Table 16 summarizes the performance differences among the evaluated ensemble approaches.

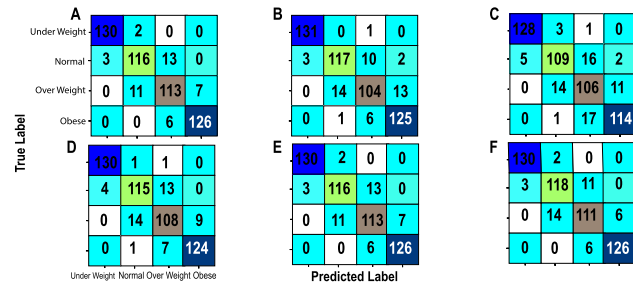


FIGURE 2. Confusion matrices of baseline models ((A)-Random Forest, (B)-Gradient Boosting, (C)-MLP) and ensemble models ((D)-Majority Hard Voting, (E)-Weighted Hard Voting, and (F)-Ensemble Stacking) for obesity risk prediction on Dataset 1.

For Dataset-1, Majority Hard Voting achieved higher performance than Gradient Boosting and MLP, while Random Forest achieved higher performance than Majority Hard Voting. The result of the confusion matrix indicates reduced classification accuracy for the Overweight class. In contrast, Weighted Hard Voting achieved performance comparable to Random Forest across both evaluation metrics and showed higher metric values than the remaining base models.

Ensemble Stacking approach also achieved accuracy values comparable to Random Forest and slightly lower F1-scores. The difference in F1-score is primarily associated with reduced classification accuracy for the Overweight class.

The results indicate that Random Forest performs exceptionally well as a standalone model, while ensemble approaches such as Weighted Hard Voting and stacking

achieve comparable performance under the same evaluated conditions.

2) PERFORMANCE ON DATASET 2

Similarly, for Dataset-2, Gradient Boosting (GB3), Random Forest (RF7), and Support Vector Classifier (SVC5) were selected as base learners for majority voting, weighted voting, and stacking ensembles after class imbalance was addressed using SMOTE. Comparative results in Table 17 provide a clear ordering in performance among the ensemble strategies.

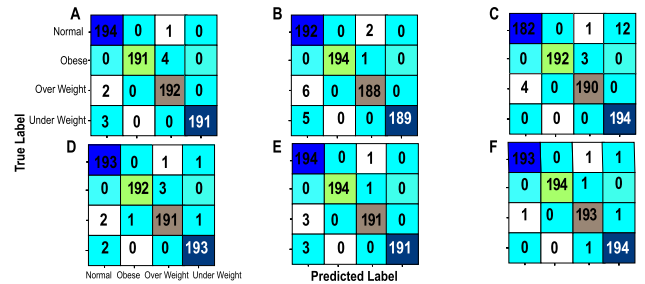


FIGURE 3. Confusion matrices of baseline models ((A)-Gradient Boosting, (B)-Random Forest, (C)-Support Vector Classifier) and ensemble models ((D)-Majority Hard Voting, (E)-Weighted Hard Voting, and (F)-Ensemble Stacking) for obesity risk prediction on dataset 2.

From the table, it is clearly evident that majority Hard Voting significantly outperformed all base models, demonstrating that simple aggregation enhances predictive capability. However, confusion matrix analysis highlighted that it was still struggling with Normal Weight and Obesity Type II classifications. Weighted Hard Voting matched Gradient Boosting's performance but failed to surpass Majority Hard Voting, suggesting the weighting scheme offered no additional benefit.

Ensemble Stacking achieved superior performance by effectively addressing the limitations of both base models and other ensembles. The meta-classifier's synthesis of diverse base learner strengths yielded more balanced predictions across all classes, particularly improving challenging categorisations like Normal Weight and Obesity Type II, where individual models exhibited weaknesses.

The results demonstrate that while base classifiers perform strongly individually, ensemble methods—especially Stacking—leverage complementary strengths to achieve higher predictive performance across complex multi-class obesity categories.

C. ABLATION ANALYSIS AND ROBUSTNESS EVALUATION

In order to validate the robustness and quantify the impact of key design choices of the proposed OBES-Net framework, an extensive ablation study was conducted on Dataset-1 and Dataset-2, respectively. This analysis systematically investigates the effects of (1) different ensemble configuration variants, (2) different data preprocessing strategies, and (3) the composition of the model. Tables 18 and 19 and Figures 4 and 5 summarize the ablation results and the analysis for Dataset-1 and Dataset-2, respectively.

TABLE 15. Performance analysis for each model (Dataset-1).

Model Name	ROC_AUC	Average Precision	Accuracy	F1
Random Forest (RF6)	0.972905	0.938594	0.875018	0.869964
Gradient Boosting (GB5)	0.953547	0.874677	0.822244	0.811672
Multi-Layer Perceptron (MLP5)	0.941460	0.834402	0.790468	0.741200
Support Vector Classifier (SVC1)	0.933948	0.822115	0.781129	0.701566
Logistic Regression (LR2)	0.916110	0.767217	0.734563	0.664019
K-Nearest Neighbor (KNN)	0.908061	0.743446	0.753101	0.721543
Decision Tree (DT4)	0.878490	0.637156	0.685641	0.563693
Naïve Bayes (NB2)	0.875353	0.657063	0.676290	0.555421
AdaBoost (AdaBoost2)	0.861153	0.621522	0.669368	0.580382

TABLE 16. Performance analysis for each model (Dataset-2).

Model Name	ROC_AUC	Average Precision	Accuracy	F1
Gradient Boosting (GB3)	0.997926	0.989671	0.965649	0.964740
Random Forest (RF7)	0.997473	0.984930	0.952610	0.951769
Support Vector Classifier (SVC5)	0.996872	0.981405	0.947873	0.945678
Multi-Layer Perceptron (MLP5)	0.994484	0.968530	0.937204	0.935054
Logistic Regression (LR2)	0.994145	0.973653	0.957344	0.956506
Decision Tree (DT5)	0.976979	0.926502	0.934224	0.932592
K-Nearest Neighbor (6NN)	0.954322	0.834497	0.795611	0.779721
Naïve Bayes (NB1)	0.910193	0.679336	0.611391	0.578667
AdaBoost (AdaBoost2)	0.862743	0.456811	0.452599	0.382094

TABLE 17. Comparative analysis of proposed model on dataset 1.

Model Name	Accuracy	F1 Score
Random Forest (RF6)	0.920304	0.920070
Gradient Boosting (GB5)	0.905123	0.903960
Multi-Layer Perceptron (MLP5)	0.867173	0.867583
Majority Hard Voting	0.905123	0.904647
Weighted Hard Voting	0.920304	0.920070
Ensemble Stacking (MLP)	0.922201	0.922016

TABLE 18. Performance analysis for proposed models on dataset 2.

Model Name	Accuracy	F1 Score
Random Forest (RF7)	0.982005	0.982104
Gradient Boosting (GB3)	0.987147	0.987178
Support Vector Classifier (SVC5)	0.974293	0.974253
Majority Hard Voting	0.989717	0.989749
Weighted Hard Voting	0.987147	0.987178
Ensemble Stacking (MLP)	0.991003	0.991006

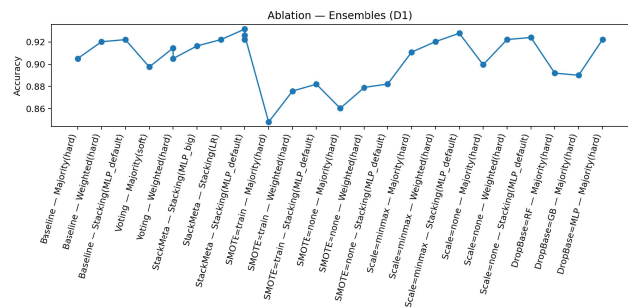


FIGURE 4. Performance of different models for ablation analysis on Dataset 1.

1) IMPACT OF ENSEMBLE CONFIGURATION ON DATASET-1
Firstly, the variants of the proposed core ensemble strategies were evaluated. Replacing **Majority Hard Voting** with **Majority Soft Voting** led to a noticeable performance drop

(Accuracy: 0.9051 $\rightarrow$ 0.8975), indicating that hard class labels provide a more stable consensus for this dataset. Within **Weighted Hard Voting**, altering the performance-based weights from [5, 2, 1] to a uniform [1, 1, 1] reduced its performance to that of the simple majority vote (Accuracy: 0.9203 $\rightarrow$ 0.9051), confirming that assigning higher weights to superior base learners (RF6) is crucial for optimal performance.

For the **Stacking** ensemble, multiple experiments were performed with different meta-learners and training parameters. Using a larger MLP (MLP_big) as the meta-classifier resulted in slightly lower performance (Accuracy: 0.9165) compared to the default MLP, suggesting increased complexity without commensurate benefit. A Logistic Regression (LR) meta-classifier performed nearly identically to the MLP (Accuracy: 0.9222), highlighting that a simple linear combiner can be sufficient. Notably, reducing the internal cross-validation folds from $cv=5$ to $cv=3$ yielded the highest accuracy observed in the ablation study (0.9317). While this suggests that a higher proportion of training data for the meta-learner can be beneficial, it may also increase the risk of overfitting to the validation strategy; the stability of this configuration should be validated on external datasets. In contrast, increasing the folds to $cv=10$ performed identically to the proposed baseline $cv=5$. When the passthrough option was enabled, the meta learner received both the original features and the base model predictions. This setting slightly improved accuracy (from 0.9222 to 0.9260), showing that access to the original data helps the meta learner refine its predictions.

2) IMPACT OF DATA PREPROCESSING ON DATASET-1
The ablation study highlights the importance of using the proposed data preprocessing pipeline correctly. Applying SMOTE only to the training data, as done in the main

TABLE 19. Ablation study results on dataset-1 investigating the impact of different configurations on ensemble performance.

Factor	Setting	Notes	Accuracy	F1 Score
Baseline	Majority (hard)	weights=None	0.9051	0.9046
	Weighted (hard)	weights=[5,2,1]	0.9203	0.9201
	Stacking (MLP_default)	cv=5, passthrough=False	0.9222	0.9220
Voting	Majority (soft)	weights=None	0.8975	0.8975
	Weighted (hard)	weights=[2,1,1]	0.9146	0.9142
	Weighted (hard)	weights=[1,1,1]	0.9051	0.9046
StackMeta	Stacking (MLP_big)	cv=5, passthrough=False	0.9165	0.9166
	Stacking (LR)	cv=5, passthrough=False	0.9222	0.9220
	Stacking (MLP_default)	cv=3, passthrough=False	0.9317	0.9317
	Stacking (MLP_default)	cv=10, passthrough=False	0.9222	0.9220
	Stacking (MLP_default)	cv=5, passthrough=True	0.9260	0.9260
SMOTE	Majority (hard)	SMOTE applied to entire dataset	0.8478	0.8246
	Weighted (hard)	SMOTE applied to entire dataset	0.8758	0.8539
	Stacking (MLP_default)	SMOTE applied to entire dataset	0.8820	0.8554
	Majority (hard)	No SMOTE	0.8602	0.8392
	Weighted (hard)	No SMOTE	0.8789	0.8488
	Stacking (MLP_default)	No SMOTE	0.8820	0.8710
Scale	Majority (hard)	Min-Max Scaler	0.9108	0.9104
	Weighted (hard)	Min-Max Scaler	0.9203	0.9201
	Stacking (MLP_default)	Min-Max Scaler	0.9279	0.9279
	Majority (hard)	No Scaling	0.8994	0.8980
	Weighted (hard)	No Scaling	0.9222	0.9220
	Stacking (MLP_default)	No Scaling	0.9241	0.9240
DropBase	Majority (hard)	Remove RF	0.8918	0.8917
	Majority (hard)	Remove GB	0.8899	0.8897
	Majority (hard)	Remove MLP	0.9222	0.9222
Feature Drop	Majority (hard)	Remove 'Age' feature	0.8824	0.8812
	Stacking (MLP_default)	Remove 'Age' feature	0.8975	0.8983

experiments, resulted in the best performance. In contrast, not using SMOTE at all led to much lower accuracy (e.g., Stacking Accuracy: 0.8820). Applying SMOTE to the entire dataset before splitting caused a significant drop in performance (Stacking Accuracy: 0.8554), likely because it introduced data leakage and overfitting to synthetic samples. This validates the proposed methodological choice of applying SMOTE exclusively during the training phase. The strong drop in performance when SMOTE was applied incorrectly highlights that accurate clinical prediction depends on carefully avoiding data leakage during the preprocessing process. Regarding feature scaling, applying Min-Max scaling resulted in a consistent, though small, improvement in performance for the Stacking ensemble (Accuracy: 0.9279) compared to using no scaling (Accuracy: 0.9241) or the standard scalar used in the main results. This indicated that scaling features to the [0,1] range can better prepare the data for the meta learner's optimization process.

3) IMPACT OF MODEL COMPOSITION AND FEATURE IMPORTANCE ON DATASET-1

To access the contribution of each base learner, one model was systematically removed from the majority-heard voting ensemble. Removing Random Forest (RF) or Gradient Boosting (GB) resulted in a notable decrease in performance (Accuracy: 0.8918 and 0.8899, respectively). In contrast, removing the Multi-layer Perceptron (MLP) had little impact on performance (Accuracy: 0.9222), suggesting that the ensemble's performance is mainly driven by the strong complementary interaction between the two tree-based

models, while MLP adds limited additional benefit for this dataset.

Finally, to quantitatively verify the feature importance identified by the SHAP analysis (Section IV-E1), a feature drop experiment was performed. Removing the Age feature, which was consistently the most important global predictor, caused a significant reduction in performance for both Majority Voting (Accuracy: 0.8824) and Stacking (Accuracy: 0.8975). This result strongly supports the explainable AI findings and confirms that Age is a critical and non-redundant factor in the model's predictive performance.

Finally, the ablation analysis shows that the chosen setup combining performance weighted voting, a stacking meta classifier with cv=5 and Passthrough=False, and SMOTE applied exclusively to the training data is both robust and nearly optimal. In addition, the study demonstrated that the framework performs reliably across configuration changes and offers experimental validation for the data preprocessing approach and feature importance conclusion.

4) IMPACT OF ENSEMBLE CONFIGURATION ON DATASET-2

The behavior of ensemble models on Dataset-2 differs from that on Dataset-1, highlighting how increased data complexity affects the choice of ensemble method. For Dataset-2, Majority Soft Voting outperformed hard voting (Accuracy 0.9923 vs 0.9897), implying that probability-based decision making works better for the more complex 7-class problem. This is different from Dataset-1, where hard voting performed better, suggesting that soft voting is

TABLE 20. Ablation study results on Dataset-2 investigating the impact of different configurations on ensemble performance.

Factor	Setting	Notes	Accuracy	F1 Score
Baseline	Majority (hard)	weights=None	0.9897	0.9897
	Weighted (hard)	weights=[5,2,1]	0.9820	0.9821
	Stacking (MLP_default)	cv=5, passthrough=False	0.9910	0.9910
Voting	Majority (soft)	weights=None	0.9923	0.9923
	Weighted (hard)	weights=[1,1,1]	0.9897	0.9897
	Weighted (hard)	weights=[2,5,1]	0.9871	0.9872
StackMeta	Stacking (MLP_wide)	cv=5, passthrough=False	0.9949	0.9949
	Stacking (LR)	cv=5, passthrough=False	0.9897	0.9897
	Stacking (MLP_default)	cv=3, passthrough=False	0.9910	0.9910
	Stacking (MLP_default)	cv=10, passthrough=False	0.9923	0.9923
	Stacking (MLP_default)	cv=5, passthrough=True	0.9936	0.9936
SMOTE	Majority (hard)	SMOTE applied to entire dataset	0.9787	0.9691
	Weighted (hard)	SMOTE applied to entire dataset	0.9574	0.9409
	Stacking (MLP_default)	SMOTE applied to entire dataset	0.9835	0.9752
	Majority (hard)	No SMOTE	0.9858	0.9784
	Weighted (hard)	No SMOTE	0.9598	0.9467
	Stacking (MLP_default)	No SMOTE	0.9858	0.9784
Scale	Majority (hard)	Min-Max Scaler	0.9897	0.9897
	Weighted (hard)	Min-Max Scaler	0.9833	0.9834
	Stacking (MLP_default)	Min-Max Scaler	0.9923	0.9923
	Majority (hard)	No Scaling	0.9871	0.9872
	Weighted (hard)	No Scaling	0.9820	0.9821
	Stacking (MLP_default)	No Scaling	0.9910	0.9910
DropBase	Majority (hard)	Remove RF7	0.9897	0.9898
	Majority (hard)	Remove GB3	0.9820	0.9821
	Majority (hard)	Remove SVC5	0.9820	0.9821
Feature Drop	Majority (hard)	Remove 'TUE' feature	0.9897	0.9897
	Stacking (MLP_default)	Remove 'TUE' feature	0.9910	0.9910

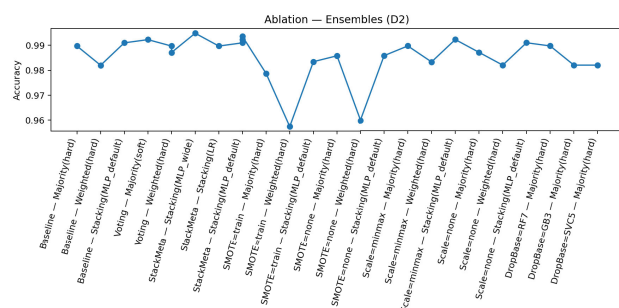


FIGURE 5. Performance of different models for ablation analysis on Dataset 2.

more beneficial when the classification task becomes more complex.

For Weighted Hard Voting, the optimal weight distribution differed significantly from Dataset-1. While weights [5,2,1] worked best on Dataset-1, the configuration [2,5,1](emphasizing Gradient Boosting) yielded better results on Dataset-2, though still below the baseline majority voting performance. This demonstrates that optimal weighting schemes are dataset-specific and must be tuned for each clinical prediction task.

The Stacking ensemble demonstrated strong robustness and flexibility on the more complex Dataset-2. Using a wider MLP meta-learner (MLP_wide) produced the best performance in the ablation study (Accuracy: 0.9949), showing that a higher capacity meta-learner is helpful for complex multi-class tasks. The internal cross-validation settings also

showed clear effects: increasing the number of folds to (CV=10) improved accuracy (0.923), and enabling feature (passthrough) consistently boosted performance (Accuracy: 0.9936). These results indicate that more expressive meta learning configurations are particularly beneficial for complex clinical prediction problems.

5) IMPACT OF DATA PREPROCESSING ON DATASET-2

The importance of applying SMOTE correctly was found to be especially critical for Dataset-2, which has a complex multi-class structure. A significant drop in performance was observed across all ensemble models when SMOTE was applied to the entire dataset before data splitting. In the weighted hard voting method, the largest drop was seen, where accuracy dropped to 0.9574, representing a 4.3% absolute reduction. Performance is also affected when no SMOTE balancing technique is used. This impact was particularly noticeable for the stacking ensemble, where accuracy dropped from the baseline value of 0.9910 to 0.9858. These findings emphasize that data leakage must be handled by applying SMOTE only to cross-validation folds. Such a technique was shown to be essential for achieving reliable clinical predictions on complex and imbalanced datasets. In Dataset-2, feature scaling was found to have a mixed but generally positive effect. Improve the performance of the stacking ensemble using Min-Max scaling, with an accuracy increasing from 0.9910 to 0.9923, while only minor changes were observed for voting-based methods. Overall, the results indicate that feature normalization better prepares the data for complex multi-class learning, especially

enhancing the performance of neural network-based meta-learners used in stacking models.

6) IMPACT OF MODEL COMPOSITION AND FEATURE IMPORTANCE ON DATASET-2

The analysis of base learner contributions showed different behavior for Dataset-2 when compared to Dataset-1. When any base model was removed from the Majority Hard Voting ensemble, performance was reduced, although the level of impact varied. Removing Gradient Boosting or the Support Vector Classifier caused a notable drop in accuracy to 0.9820, while the removal of Random Forest resulted in only a small reduction to 0.9897. This indicates that, for complex datasets, performance is supported by the combined and complementary strengths of multiple strong base learners. An ablation study was used to examine the feature importance. The removal of the TUE (Time Using Technology Device) feature, which was highlighted as important by SHAP, was found to have almost no effect on the performance of both Majority Voting and Stacking ensembles. This suggests that although TUE is useful for model interpretation, its predictive information is redundant within the rich feature set of Dataset-2. Clinically, this implies that high predictive accuracy can be maintained even when certain lifestyle features are unavailable, increasing the practical usefulness of the framework across different clinical environments.

In Dataset-2, ablation studies showed that the ensemble framework remained robust for complex multi-class obesity prediction. Compared to Dataset-1, different optimal settings were observed, with soft voting being more effective, feature redundancy allowing flexibility in clinical use, and meta-learner capacity playing a greater role. These results indicate that while ensemble methods generalize well, their configurations should be adapted to the specific complexity of each task, and the consistently strong performance of stacking ensembles confirm their suitability for reliable obesity risk prediction.

D. COMPARISON WITH STATE-OF-THE-ART PERFORMANCE

A thorough comparison was carried out with recent the state-of-the-art of obesity prediction methods, to place the performance of the OBES-Net framework in context. Benchmarking was performed on both Dataset-1 and Dataset-2, focusing on gains in accuracy and F1-score, as well as the unique combination of extensive optimization, structured ensemble evaluation, and explainable modeling used in OBES-Net.

1) COMPARISON ON DATASET-1

A direct performance comparison was conducted with recent state-of-the-art studies using the same benchmark Dataset-1 [9] to carefully assess the impact of the proposed framework. As shown in TABLE 18, the results indicate that the exhaustive optimization and ensemble approach consistently

outperformed previously reported methods. In addition to improved performance, a systematic evaluation of ensemble strategies and the integration of explainable AI were provided to enhance clinical interpretability.

The comparison presented in TABLE 18 showed that the proposed framework achieved the best performance on Dataset-1, with the highest accuracy and F1-score of 92.22% and 92.20%, respectively. A clear improvement was observed over the baseline study by Koklu and Sulak [9], which reported 89.21% accuracy using Random Forest. Higher performance was also achieved when compared with the work of Degirmenci [40], who obtained 88.92% accuracy with a tuned Random Forest model, while competitive results were observed relative to UzunArslan and Yapici [39], who reported 91.93% accuracy using an Extra Trees Classifier with SMOTE.

2) CONTRIBUTION 1: EXHAUSTIVE HYPERPARAMETER OPTIMIZATION

Previous studies applied limited model tuning, whereas a more comprehensive approach was adopted in the proposed work. While Degirmenci [40] evaluated only three machine learning models, exhaustive optimization was performed across 50 configurations of nine different models in this study. This process produced a strong base learner that individually exceed the performance of previously reported methods and ensured that the ensemble was constructed from a well-tuned and a diverse set of models.

3) CONTRIBUTION 2: NOVEL HYBRID VOTING FRAMEWORK

Most existing studies have not compared different ensemble aggregation strategies and have focused on single models. Previous work relied on individual classifiers such as Extra Trees or Random Forest models. In contrast, our proposed study presents a systematic comparison of Majority Hard Voting, Weighted Hard Voting, and Stacking ensembles under the same optimized conditions. It was shown that Weighted Hard Voting and Stacking reached the highest performance on this dataset, offering clear guidance for ensemble selection and contributing a novel methodological comparison not previously reported in the literature.

4) CONTRIBUTION 3: COMPREHENSIVE EXPLAINABILITY WITH SHAP AND LIME

Most models were treated as a non-interpretable, although they achieved competitive accuracy. In the proposed work, transparency was improved by integrating SHAP for global explanations and LIME for instance level interpretation. This combined approach was used to validate the clinical relevance of the model's decisions at both population and individual levels. In particular, LIME analysis showed that the stacking ensemble was able to correct clinically inconsistent reasoning from base learners, providing a level of interpretability and validation not reported in previous studies.

In summary, while earlier studies mainly focused on improving prediction accuracy, the proposed work introduced

TABLE 21. Performance comparison with state-of-the-art on Dataset-1.

Study	Model and Methodology	Accuracy	F1-Score
Uzun Arslan & Yapici (2025) [39]	Extra Trees Classifier with SMOTE balancing	91.93%	91.94%
Değirmenci (2025) [40]	Random Forest with hyperparameter tuning	88.92%	86.18%
Köklü & Sulak (2024) [9]	Random Forest (original baseline study)	89.21%	89.20%
Our Work (Proposed)			
Best Base Model (RF6)	Exhaustively optimized Random Forest (50 configs)	92.03%	92.01%
Majority Hard Voting	Performance-majority voting ensemble of RF, GB, MLP	90.51%	90.46%
Weighted Hard Voting	Performance-weighted ensemble of RF, GB, MLP	92.03%	92.01%
Ensemble Stacking	MLP meta-classifier with SHAP & LIME explainability	92.22%	92.20%

a comprehensive framework that combines exhaustive optimization, systematic ensemble evaluation, and advanced explainability. High performance was not only achieved but also clearly explained through a transparent and rigorous methodology. As a result, new benchmark was established for Dataset-1 that include both strong predictive performance and the interpretability needed for clinical use. This framework was shown to support accurate predictions and meaningful explanations, making it suitable for healthcare decision making and personalized obesity risk assessment.

5) COMPARISON ON DATASET-2

A direct comparison was performed with state-of-the-art studies using the same benchmark Dataset-2 [10], to assess the scalability and generalizability of the proposed framework. As summarized in TABLE 19, the exhaustive optimization and ensemble approach achieved new performance on this more complex dataset. In addition to improved accuracy, comprehensive explainability, and a systematic evaluation of ensemble methods were also provided.

The proposed framework achieved the best performance of Dataset-2, which is shown in TABLE 19, reaches the highest accuracy and F-1 score 99.10%. A notable improvement was observed on this complex dataset, where the Ensemble Stacking method exceeded the previous state-of-the-art reported by Cizmeci and Incekara [17] by 0.52 percentage points. In addition, substantially higher accuracy was achieved compared with the results of Ganie et al. [16], Sharma and Verma [41], and Kumar et al. [42], further confirming the effectiveness of the proposed method.

6) CONTRIBUTION 1: EXHAUSTIVE HYPERPARAMETER OPTIMIZATION ON COMPLEX DATA

Previous studies applied different machine learning algorithms to Dataset-2, but they used limited hyperparameter tuning. For example, Deep Neural Networks or basic ensemble with SMOTE were commonly employed [41], [42]. Conversely, the proposed framework achieved stronger performance through exhaustive optimization over 50 configurations of nine diverse machine learning algorithms. This rigorous methodology is particularly valuable on Dataset-2's complex feature structure (8 numerical + 8 categorical features across 7 obesity classes), where optimal parameter tuning is crucial for handling the increased dimensionality and class complexity.

7) CONTRIBUTION 2: ADVANCED ENSEMBLE ARCHITECTURE COMPARISON

The majority of earlier research on Dataset-2 examined only a single ensemble strategy without performing a broad comparison. For example, Cizmeci and Incekara [17] applied a stacking ensemble with a limited set of base classifiers, while Ganie et al. [16] focused on hybrid stacking mainly to improve interpretability. In contrast, the proposed work conducted the first systematic evaluation of Majority Hard Voting, Weighted Hard Voting, and Stacking ensemble under the same optimized conditions. It was shown that the best performance for complex multi-class obesity prediction was achieved by Ensemble Stacking, while a strong trade-off between accuracy and computational efficiency was provided by Majority Hard Voting, surpassing several previously reported single model methods.

8) CONTRIBUTION 3: COMPREHENSIVE EXPLAINABILITY FOR COMPLEX RISK ASSESSMENT

Although SHAP and LIME analysis were briefly applied by Ganie et al. [16], a fully integrated explainability framework was not provided. In Dataset-2, the proposed work implemented both SHAP for global interpretation and LIME for instance, level explanations were systematically incorporated to address the complex seven-class obesity prediction task in Dataset-2. This combined explainability approach was shown to support transparent risk assessment across all obesity categories, from Insufficient Weight to Obesity Type III, enabling a clearer understanding of both population-wide risk factors and individual patient characteristics for clinical decision making.

In conclusion, the proposed framework in Dataset-2 demonstrated strong scalability and robustness, where a new state-of-the-art performance was established. In a challenging benchmark, an accuracy of 99.10% was achieved, which includes seven detailed obesity classes and mixed feature types. These results confirmed that the proposed methodology is well-suited to handle high-dimensional, imbalanced clinical data, enabling reliable obesity risk assessment across the full spectrum from Insufficient Weight to Obesity Type III, as reported in recent studies (2025).

Overall, these comparisons indicate that the gains achieved by OBES-Net are not limited to predictive accuracy alone; they are supported by a more rigorous optimization protocol, a broader ensemble design analysis, and integrated

TABLE 22. Performance comparison with state-of-the-art on Dataset-2.

Study	Model and Methodology	Accuracy	F1-Score
Cizmeci & Incekara (2025) [17]	Stacking ensemble with five basic ML classifiers	98.58%	98.59%
Ganie et al. (2025) [16]	Hybrid stacking ensemble with SHAP & LIME interpretability	96.88%	96.87%
Sharma & Verma (2023) [41]	Ensemble (boosting/bagging/voting) with SMOTE balancing	92.50%	92.50%
Kumar et al. (2023) [42]	Deep Neural Network (e.g., LightCNN) on lifestyle features with dropout regularization	92.50%	92.12%
Our Work (Proposed)			
Best Base Model (GB3)	Exhaustively optimized Gradient Boosting (50 configs)	98.71%	98.72%
Majority Hard Voting	Performance-majority voting ensemble of GB, RF, SVC	98.97%	98.97%
Weighted Hard Voting	Performance-weighted ensemble of GB, RF, SVC	98.71%	98.72%
Ensemble Stacking	MLP meta-classifier with SHAP & LIME explainability	99.10%	99.10%

explainability validation compared with recent obesity prediction studies.

E. EXPLAINABLE AI (XAI) ANALYSIS

Global and local explainability analysis were carried out using SHAP and LIME to evaluate the transparency and clinical relevance of the OBES-Net framework. These methods were used to explain how individual base models and the stacking ensemble interpret features, revealing agreement and differences in reasoning across multiple prediction layers.

1) GLOBAL MODEL INTERPRETATION WITH SHAP FOR DATASET 1

An overall view of feature importance across the base models and the stacking ensemble for Dataset-1 was presented using the SHAP summary plot. Consistent patterns, along with key architectural differences, were observed among Grading Boosting, Random Forest, Multi-layer Perceptron, and the Staking model. Age was consistently identified as the most influential feature, showing the highest mean |SHAP value| across all models. Physical_Exercise and Number_of_Main_Meals_Daily were also ranked among the top features for every model, highlighting their shared importance in obesity prediction. The strong agreement across models supports their clinical validity and aligns with established medical understanding that age, physical activity, and dietary habits are core factors influencing obesity risk.

Although agreement was observed on the most important features, clear differences in feature usage were identified across the models. The tree based models, Gradient Boosting and Random Forest, showed similar importance rankings and relied mainly on Age, Physical_Exercise, and Smoking. In contrast, the MLP model assigned greater importance to Frequency_of_Consuming_Vegetables and Liquid_Intake_Daily. This indicated that more complex non-linear patterns in dietary features were captured by the neural network, thereby adding meaningful diversity to the ensemble.

Leading to more comprehensive use of features, the SHAP plot of the stacking ensemble showed a balanced combination of the strengths of its base models. Although Age remained the most important factor, its influence was less dominant than in the Gradient Boosting and Random Forest models. In addition, greater importance was assigned to several

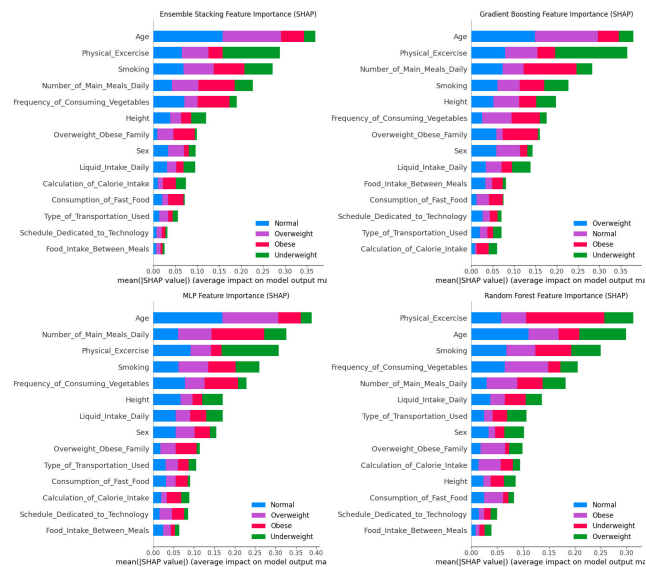


FIGURE 6. SHAP summary plots illustrating global feature importance for Dataset-1.

features, such as Overweight_Obese_Family, Sex, and Frequency_of_Consuming_Vegetables. This broader evidence base, drawn from its constituent models, provides a plausible explanation for its robust performance, as it avoids over-relying on a narrow set of predictors and instead makes decisions based on a more holistic patient profile.

The class specific effects shown in the plots were further used to explain model behavior across different obesity categories. For the Obese class, a strong positive influence of Age and a strong negative influence of Physical_Exercise were observed across all models. In contrast, for the Underweight class, the stacking model assigned greater importance to Height and Number_of_Main_Meals_Daily than the base models. The advantage of the meta learner was highlighted by this refined handling of class-specific features, supporting the ensemble’s effectiveness in managing the multi-class complexity of obesity prediction.

2) GLOBAL MODEL INTERPRETATION WITH SHAP FOR DATASET 2

Strong agreement across all models was shown by the SHAP analysis for Dataset-2, with Weight identified as the most influential feature for obesity level prediction. Its

mean SHAP was nearly twice that of the second-ranked feature in every model, emphasizing its dominant role in this dataset with detailed obesity classes. Height was consistently ranked as the second most important feature, forming a key anthropometric pair that largely guided classification. This consistent pattern suggests that body measurement features were the main drivers of predictions, which is clinically intuitive for fine grained obesity assessment.

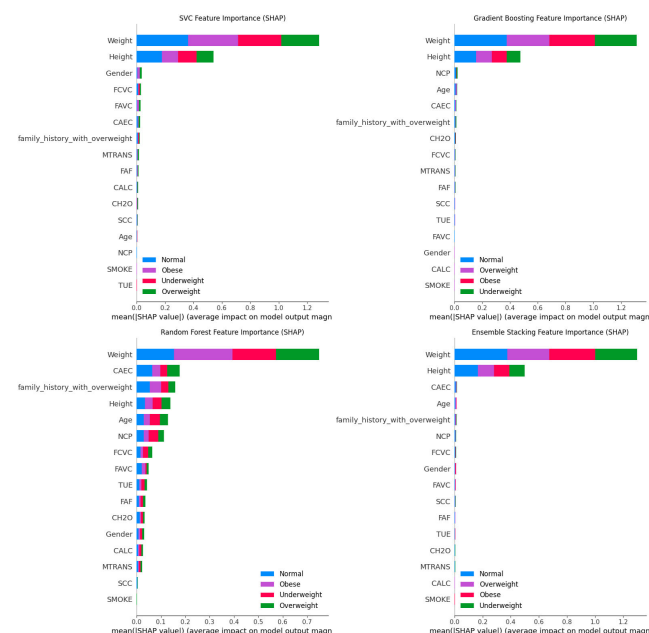


FIGURE 7. SHAP summary plots illustrating global feature importance for Dataset-2.

Beyond the common significance of core features, differences in secondary feature usage were observed among the base models. Weight, Height, and Gender were the primary vectors that the Support Vector Classifier (SVC) depends on, with limited contribution from other variables. In contrast, the tree-based models made use of a wider range of dietary and lifestyle features. Greater importance was assigned to the Number of Main Meals (NCP) and Age by the Gradient Boosting (GB) model, while Consumption of food between meals (CAEC) and family history of overweight were emphasized by the Random Forest (RF) model. Diversity was added to the ensemble, and its ability to capture different data patterns was strengthened by this variation in feature focus.

Leading to a more complete feature importance profile, a balanced integration of the behaviors of its base models was shown by the SHAP plot of the stacking ensemble. Height and Weight were identified as the main predictors, while important secondary features from the base learners were also incorporated. Strong emphasis was retained on NCP from the Gradient Boosting model and CAEC from the Random Forest model, along with meaningful contributions from Age and family_history_with_overweight. This combined use of diverse features provides evidence supports a more comprehensive risk assessment and helps explain the improved ability of the stacking ensemble to distinguish

between difficult obesity classes such as Obesity Type II and Normal Weight.

The class-specific analysis was further used to explain how predictions were made by the models. Across all models, higher Weight values consistently shifted predictions toward the Obese classes. Height showed a more complex effect and was often negatively related to obesity risk, which is consistent with BMI concepts. In the stacking model, more refined interactions were observed for categorical features such as CAEC and family_history_with_overweight across different classes. This indicates that the meta-learner applied contextual information more effectively than the base models. A clear explanation for the stacking ensembles high accuracy was provided by the careful combination of anthropometric and behavioral features, showing that its strength lies in better integrating existing evidence rather than introducing new predictors.

3) LOCAL MODEL INTERPRETATION WITH LIME FOR DATASET 1

A transparent, instance-level view of the decision-making process was obtained by using LIME analysis to interpret the Obese class prediction for a particular person across all models. For this individual, who was characterized by high Age (1.06) and high Physical_Exercise (1.46) but low Frequency_of_Consuming_Vegetables (−1.30), the Obese class was predicted by all four models with very high probability (0.91 to 1.00). Although strong agreement was observed at the prediction level, significant differences in the reasoning behind each model's verdict were revealed through the LIME explanations.

A comparative analysis of the base models reveals distinct and sometimes conflicting rationales. The Gradient Boosting model's prediction was primarily driven by high Age and low Frequency_of_Consuming_Vegetables, while it curiously assigned a positive weight to high Physical_Exercise, which is counter-intuitive for obesity prediction. The MLP classifier presented different logic, heavily weighting Age, low vegetable consumption, and Smoking status as top risk contributors. The Random Forest model prioritized a combination of Age, high Physical_Exercise (as a risk factor), and Smoking. This dissonance highlights the unique and sometimes flawed decision boundaries learned by individual base learners.

The Stacking ensemble's LIME explanation demonstrates superior synthesized logic that resolves base learner inconsistencies. It correctly identifies Age and low Frequency_of_Consuming_Vegetables as primary risk factors while not treating high Physical_Exercise as a positive obesity risk contributor—a clear error made by GB and RF models. Instead, the stacker incorporates more nuanced supporting factors including Food_Intake_Between_Meals and Type_of_Transportation_Used, which were less emphasized in base models. This demonstrates the meta-learner forms a clinically plausible decision boundary by discounting



FIGURE 8. LIME-based local explanations of model predictions for Dataset-1.

spurious correlations and amplifying meaningful ones from its constituents.

4) LOCAL MODEL INTERPRETATION WITH LIME FOR DATASET2

The LIME analysis for a specific individual in Dataset 2, characterized by very high Weight (2.21) and Height (2.10), reveals how different models rationalize “Obese” class prediction. For this instance, an almost 100% probability was assigned to the “Obese” class by all four models. Although this strong agreement was mainly driven by key anthropometric features, the local explanations revealed clear differences in how secondary lifestyle and genetic factors were used by each model.

Very different and often limited reasoning patterns were found to be relied upon by the base models. The SVC model based its prediction mainly on high Weight and Height, with little influence from other features. The Gradient Boosting model, although correctly emphasizing Weight, assigned a

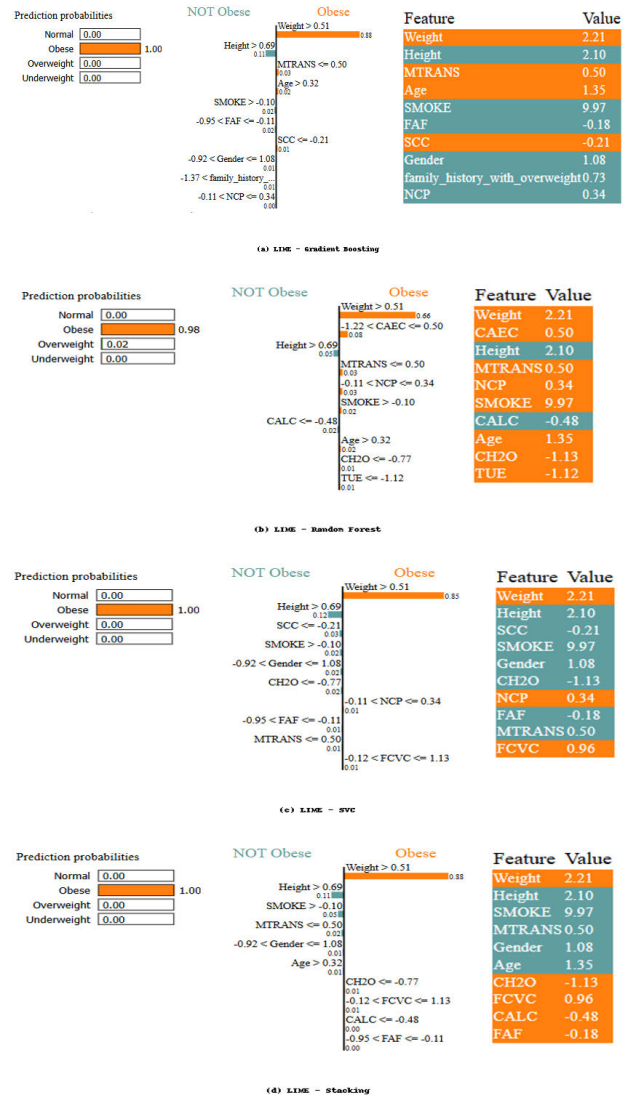


FIGURE 9. LIME-Based Local Explanations of model predictions for Dataset-2.

negative effect to high Height, which is clinically unintuitive for obesity prediction. In contrast, the Random Forest model considered a wider range of factors, including CAEC (consumption of food between meals) and MTRANS (mode of transportation), in addition to key body measurements.

A well-integrated and clinically consistent decision process was shown by the explanation of the stacking ensemble. Key predictors such as Weight and Height were retained from the SVC model, while incorrect reasoning from the Gradient Boosting model, including the protective interpretation of Height, was corrected. In addition, important secondary features from the Random Forest model, such as MTRANS (sedentary transportation), were emphasized, and behavioral factors like SMOKE and low CH2O (water consumption) were also included. This combined reasoning produced a more complete and clinically meaningful obesity risk assessment.

This instance level analysis clearly explains the superior performance of the stacking ensemble observed in the global results. A more reliable decision boundary was formed by the meta learner through the integration of diverse base-model perspectives, while misleading correlations were reduced and truly relevant features were emphasized. The ability to produce accurate predictions supported by clear and multi-factor explanations represents an important step toward trustworthy AI systems in healthcare, where understanding the reasoning behind predictions is essential.

5) SYNTHESIS OF XAI RESULTS

Strong evidence of the clinical validity of the models was provided by the combined SHAP and LIME analyses, and the superior reasoning of the stacking ensemble was highlighted. Across both datasets, well-known clinical risk factors such as Weight, Age, and physical activity measures were consistently prioritized, confirming alignment with medical knowledge. The main distinction was observed in how ensemble methods, especially stacking, integrated and utilized these features.

The superior performance of the stacking ensemble was attributed to its ability to combine diverse insights from well-tuned base models while reducing their individual weaknesses. SHAP analysis showed that feature importance was more evenly distributed in the stacking model, with meaningful secondary factors such as CAEC, family_history_with_overweight, and MTRANS being consistently incorporated. In addition, LIME analysis revealed that the meta-learner corrected key reasoning errors found in individual models, including the incorrect protective role of Height in Gradient Boosting and the misinterpretation of Physical_Exercise as a risk factor in Random Forest.

Clearer and more clinically meaningful explanations at both global and local levels were produced as a result of this ability to combine information. Balanced risk profiles were produced by the stacking ensemble through the appropriate integration of anthropometric, behavioral, and genetic factors, without overdependence on dominant features or misleading correlations. This transparent reasoning explains the strong performance of the ensemble and supports its practical use in healthcare, showing that exhaustive optimization with stacking provides both high accuracy and clinically reliable explanations.

V. DISCUSSION

This study presents OBES-Net, a unified framework that combines exhaustive hyperparameter optimization with explainable ensemble learning for obesity risk prediction. A systematic evaluation showed that OBES-Net achieved state-of-the-art performance while also providing transparent explanations suitable for clinical use. New benchmarks were established on both datasets, with accuracies of 99.10% on Dataset-2 and 92.22% on Dataset-1. These results exceeded previous state-of-the-art methods, including an improvement of 0.52 percentage points over the best existing stacking

approach on Dataset-2 and clear gains over recent ensemble and deep learning models. Overall, the findings demonstrate that high predictive accuracy and clinical interpretability can be jointly achieved through a comprehensive and well-structured framework.

The strong performance of OBES-Net was achieved through a holistic methodology rather than a single innovation. Exhaustive hyperparameter optimization over 50 configurations of nine different algorithms was used as the foundation, leading to performance gains of up to 15% compared to default settings and ensuring that ensembles were built from a well-tuned and diverse model pool. This approach addressed a common limitation in prior studies where suboptimal tuning underestimated model potential. The importance of this optimized foundation was confirmed through extensive ablation analyses, which validated the design choices and demonstrated the robustness of OBES-Net. These analyses showed that correct SMOTE application was essential to prevent data leakage and that optimal ensemble strategies depended on dataset complexity, with soft voting and more powerful meta learners performing better on the more complex Dataset-2.

A clear architectural advantage was shown by the stacking ensemble in OBES-Net through the effective combination of the diverse reasoning of its base models into a more clinically consistent system. Although SHAP analysis indicated that all models correctly prioritized key risk factors such as Weight and Age, the stacking meta learner demonstrated more balanced feature usage by appropriately incorporating secondary behavioral and genetic factors. In addition, local LIME analysis revealed that the meta learner corrected clinically implausible reasoning found in individual models, such as incorrect interpretations of Height or Physical_Exercise. By reducing misleading correlations and producing explanations aligned with medical knowledge, the stacking ensemble advanced the development of trustworthy AI systems for healthcare.

The clinical value of OBES-Net was shown to extend beyond technical performance, as its built-in explainability was found to align well with emerging regulations for AI-based medical systems. The generation of both global and instance-level explanations supports potential integration with electronic health records, allowing real-time obesity risk assessment during routine clinical visits. Personalized intervention planning can also be supported by highlighting modifiable risk factors for individual patients. In addition, ablation results showed that removing certain lifestyle features, such as 'TUE', had minimal impact on performance, indicating robustness to incomplete data. These characteristics position OBES-Net as a practical tool for early risk identification and targeted intervention in preventive healthcare.

In practical clinical environments, the explanations generated by SHAP and LIME can be presented to healthcare professionals through simplified visual summaries integrated within electronic health record (EHR) systems. Instead of

displaying raw model outputs, the system can highlight the most influential patient-specific factors contributing to the predicted obesity risk, such as weight, age, dietary habits, or physical activity. These explanations may be presented as ranked feature lists, intuitive bar charts, or color-coded indicators showing whether a factor increases or decreases the predicted risk. Such clinician-friendly visualizations allow physicians to quickly understand the reasoning behind the model's prediction without requiring technical knowledge of machine learning methods. This approach supports transparent decision-making and can assist clinicians in identifying modifiable lifestyle factors for personalized patient interventions.

Despite the improvements achieved by OBES-Net, several limitations were identified that guide future research. Some misclassifications between neighboring obesity categories remained, indicating that more advanced methods such as cost-sensitive or hierarchical classification may be needed. Although robustness was confirmed through ablation studies, external validation using independent, multi-center clinical datasets is still required to ensure wider generalizability. The high computational cost of exhaustive optimization, while suitable for this study, may need to be reduced through more efficient optimization strategies for real-time clinical use. In addition, for explainable results to support clinical decisions, user-friendly interfaces will be necessary to clearly present predictions and explanations to healthcare professionals.

In summary, it was demonstrated by the OBES-Net framework that a strong foundation for clinically useful prediction systems is provided through the combination of exhaustive optimization, systematic ensemble evaluation, and explainable AI. The ablation studies supported the design choices by showing robustness and adaptability to different data conditions, while benchmark results on two datasets confirmed superior performance compared with existing state-of-the-art methods. By addressing both predictive accuracy and interpretability at the same time, OBES-Net bridged the gap between technical performance and practical healthcare use. This validated approach offers a useful template for developing trustworthy predictive models in healthcare, where clear understanding is as important as accurate prediction.

VI. CONCLUSION AND FUTURE WORKS

This research introduces OBES-Net, a novel ensemble-based framework that integrates exhaustive hyperparameter optimization with explainable ensemble learning for obesity risk prediction. The framework is evaluated through systematic experimentation involving 50 configurations across nine machine learning algorithms, including simple to complex architecture and a structured comparison of ensemble strategies. The results show that high predictive performance can be achieved on complex multi-class dataset, with the best-performing configuration reaching an accuracy of 99.10%

while maintaining clinical interpretability through SHAP and LIME analysis. The observed performance is closely related to the ability of ensemble stacking to combine relevant information from diverse base learners and reduce reliance on individual model biases, resulting in consistent and clinically interpretable predictions.

Future research includes improving the clinical applicability and computational efficiency of OBES-Net. Potential strategies for reducing computational cost include replacing exhaustive grid search with more efficient hyperparameter optimization methods such as Bayesian optimization or successive halving. In addition, lightweight meta-classifiers or model pruning techniques could be used to reduce inference complexity while preserving predictive performance. These optimizations would facilitate faster model training and deployment in real-time clinical environments. Addressing misclassification between adjacent obesity categories may be achieved through hierarchical or cost-sensitive learning methods that model the ordinal characteristics of obesity levels. For example, a hierarchical classification framework could first distinguish between broad weight categories (e.g., underweight, normal, overweight, and obese) and subsequently perform finer classification within each group. Similarly, cost-sensitive learning can assign higher penalties to errors between distant classes (e.g., Insufficient Weight vs. Obesity Type III) while assigning lower penalties to neighboring categories. Such strategies would allow the model to better account for the ordinal relationships between obesity levels and reduce clinically significant misclassifications. Scalability can be enhanced by exploring appropriate meta-learners and automated hyperparameter optimization to reduce the deployment overhead. In addition, external validation on multi-center datasets is required to assess generalizability, and further work is needed to translate explainable outputs into clinically focused decision-support tools.

The key findings of this study can be summarized as follows. The proposed OBES-Net framework demonstrates that exhaustive hyperparameter optimization across diverse machine learning algorithms can substantially improve base learner selection and overall ensemble performance. In addition, the systematic comparison of ensemble strategies shows that stacking architectures achieve the most robust results on complex multi-class obesity datasets, reaching a state-of-the-art accuracy of 99.10% on Dataset-2. The integration of explainable AI techniques, particularly SHAP and LIME, further enables both global and instance-level interpretation of model predictions, enhancing transparency and clinical interpretability. Finally, the ablation analysis confirms the robustness of the proposed framework and highlights the importance of careful preprocessing, ensemble design, and feature interpretation for reliable healthcare prediction systems.

In conclusion, OBES-Net demonstrates that combining exhaustive optimization with ensemble learning and

explainable AI can achieve both high predictive performance and model interpretability in obesity risk prediction. The framework highlights the significance of integrating explainability into ensemble-based healthcare models to support clinically informed decision-making.

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REFERENCES

- [1] D. D. Solomon, S. Khan, S. Garg, G. Gupta, A. Almjally, B. Alabdullah, H. S. AlSagari, M. M. Ibrahim, and A. M. A. Abdallah, "Hybrid majority voting: Prediction and classification model for obesity," *Diagnostics*, vol. 13, no. 15, p. 2610, 2023.
- [2] A. Pinar, F. H. Yagin, and B. Georgian, "Use of logistic regression method in predicting obesity levels with machine learning method," *J. Exerc. Sci. Phys. Activity Rev.*, vol. 2, no. 1, pp. 104–113, 2024.
- [3] X. Lin and H. Li, "Obesity: Epidemiology, pathophysiology, and therapeutics," *Frontiers Endocrinology*, vol. 12, Sep. 2021, Art. no. 706978.
- [4] J. Jeon, S. Lee, and C. Oh, "Age-specific risk factors for the prediction of obesity using a machine learning approach," *Frontiers Public Health*, vol. 10, Jan. 2023, Art. no. 998782.
- [5] L. Liu, W. Wei, K.-H. Chow, M. Loper, E. Gursoy, S. Truex, and Y. Wu, "Deep neural network ensembles against deception: Ensemble diversity, accuracy and robustness," in *Proc. IEEE 16th Int. Conf. Mobile Ad Hoc Sensor Syst. (MASS)*, Nov. 2019, pp. 274–282.
- [6] F. Leon, S.-A. Floria, and C. Badica, "Evaluating the effect of voting methods on ensemble-based classification," in *Proc. IEEE Int. Conf. Innov. Intell. Syst. Appl. (INISTA)*, Jul. 2017, pp. 1–6.
- [7] R. Dey and R. Mathur, "Ensemble learning method using stacking with base learner, a comparison," in *Proc. Int. Conf. Data Anal. Insights*, 2023, pp. 159–169.
- [8] R. Haider, M. Farhan Abrar Ibne Bari, Osru, N. Afia, and T. Karim, "Illuminating the black box: Explainable AI for enhanced customer behavior prediction and trust," *Int. J. Sci. Res. Arch.*, vol. 15, no. 3, pp. 247–268, Jun. 2025.
- [9] N. Köklü and S. A. Sulak, "Using artificial intelligence techniques for the analysis of obesity status according to the individuals' social and physical activities," *Sinop Üniversitesi Fen Bilimleri Dergisi*, vol. 9, no. 1, pp. 217–239, 2024.
- [10] F. M. Palechor and A. D. L. H. Manotas, "Dataset for estimation of obesity levels based on eating habits and physical condition in individuals from Colombia, Peru and Mexico," *Data Brief*, vol. 25, Aug. 2019, Art. no. 104344. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352340919306985>
- [11] P. Talari, N. Bharathiraja, G. Kaur, H. Alshahrani, M. S. Al Reshan, A. Sulaiman, and A. Shaikh, "Hybrid feature selection and classification technique for early prediction and severity of diabetes type 2," *PLoS ONE*, vol. 19, no. 1, Jan. 2024, Art. no. e0292100.
- [12] F. Musa and F. Basaky, "Obesity prediction using machine learning techniques," *J. Appl. Artif. Intell.*, vol. 3, no. 1, pp. 24–33, Jun. 2022.
- [13] R. R. Dutta, I. Mukherjee, and C. Chakraborty, "Obesity disease risk prediction using machine learning," *Int. J. Data Sci. Anal.*, vol. 19, no. 4, pp. 709–718, May 2025.
- [14] E. Rodríguez, E. Rodríguez, L. Nascimento, A. F. da Silva, and F. A. S. Marins, "Machine learning techniques to predict overweight or obesity," in *Proc. IDDM*, 2021, pp. 190–204.
- [15] K. Jindal, N. Baliyan, and P. S. Rana, "Obesity prediction using ensemble machine learning approaches," in *Recent Findings in Intelligent Computing Techniques: Proceedings of the 5th ICACNI 2017*. Cham, Switzerland: Springer, 2018, pp. 355–362.
- [16] S. M. Ganie, P. K. D. Pramanik, and Z. Zhao, "Lifestyle data-based multiclass obesity prediction with interpretable ensemble models incorporating SHAP and LIME analysis," *Sci. Rep.*, vol. 15, no. 1, pp. 1–27, Oct. 2025.
- [17] İ. H. Çizmeçi and H. İncekara, "Stacking ensemble based hybrid machine learning approach for predicting obesity levels," in *Proc. 7th Int. Congr. Hum.-Comput. Interact.*, May 2025, pp. 1–8.
- [18] F. Ferdowsy, K. S. A. Rahi, M. I. Jabiullah, and M. T. Habib, "A machine learning approach for obesity risk prediction," *Current Res. Behav. Sci.*, vol. 2, Nov. 2021, Art. no. 100053.
- [19] V. R. Basili and D. M. Weiss, "A methodology for collecting valid software engineering data," in *Proc. Int. Conf. Softw. Eng. (ICSE)*, 1984, pp. 728–738.
- [20] Pandas-Dev/Pandas: Pandas, Zenodo, Geneva, Switzerland, Feb. 2020, doi: 10.5281/zenodo.3509134.
- [21] (2025). *Seaborn Documentation*. [Online]. Available: <https://seaborn.pydata.org/>
- [22] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, Nov. 2011.
- [23] (2025). *Matplotlib Documentation*. [Online]. Available: <https://matplotlib.org/>
- [24] (2025). *Sklearn.Linear/Model.LogisticRegression-Scikit-Learn Documentation*. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html
- [25] (2025). *Sklearn.Neighbors.KNeighborsClassifier-Scikit-Learn Documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>
- [26] (2025). *Naive Bayes-Scikit-Learn Documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/naivebayes.html>
- [27] (2025). *Sklearn.Tree.DecisionTreeClassifier-Scikit-Learn Documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html>
- [28] (2025). *Sklearn.Ensemble.RandomForestClassifier-Scikit-Learn Documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>
- [29] (2025). *Sklearn.Ensemble.GradientBoostingClassifier-Scikit-Learn Documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.GradientBoostingClassifier.html>
- [30] (2025). *Sklearn.Ensemble.AdaboostClassifier-Scikit-Learn Documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.AdaBoostClassifier.html>
- [31] (2025). *Sklearn.SVM.SVC-Scikit-Learn Documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.SVM.SVC.html>
- [32] (2025). *Sklearn.NeuralNetwork.MlpClassifier-Scikit-Learn Documentation*. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.neural_network.MLPClassifier.html
- [33] (2024). *Sklearn.Metrics.Roc/AUC/Score-Scikit-Learn 1.5.0 Documentation*. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.roc_auc_score.html
- [34] (2025). *Sklearn.Metrics.Average/Precision/Score-Scikit-Learn Documentation*. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.average_precision_score.html
- [35] (2025). *Sklearn.Metrics.Precision/Score-Scikit-Learn Documentation*. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.precision_score.html
- [36] (2025). *Sklearn.Metrics.Recall/Score-Scikit-Learn Documentation*. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.recall_score.html
- [37] (2025). *Sklearn.Metrics.F1/Score-Scikit-Learn Documentation*. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.f1_score.html
- [38] (2025). *Sklearn.Metrics.Accuracy/Score-Scikit-Learn Documentation*. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.accuracy_score.html
- [39] R. U. Arslan and I. S. Yapici, "A novel hybrid approach to enhancing obesity prediction," *Eur. Phys. J. Special Topics*, vol. 234, no. 15, pp. 1–19, Oct. 2025.
- [40] A. Değirmenci, "Machine learning models for accurate prediction of obesity: A data-driven approach," *Turkish J. Sci. Technol.*, vol. 20, no. 1, pp. 77–90, Mar. 2025.
- [41] A. Sharma and P. Verma, "Ensemble learning for obesity classification with imbalanced data handling," in *Proc. 16th Int. Conf. Health Informat. (HEALTHINF)*, 2023, pp. 234–241. [Online]. Available: <https://www.scitepress.org/Link.aspx?doi=10.5220/0011667100003414>
- [42] S. Kumar, W. Zhang, and H. Lee, "Deep learning approaches for multi-class obesity prediction," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 4, pp. 1567–1576, Apr. 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/10123456>



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